

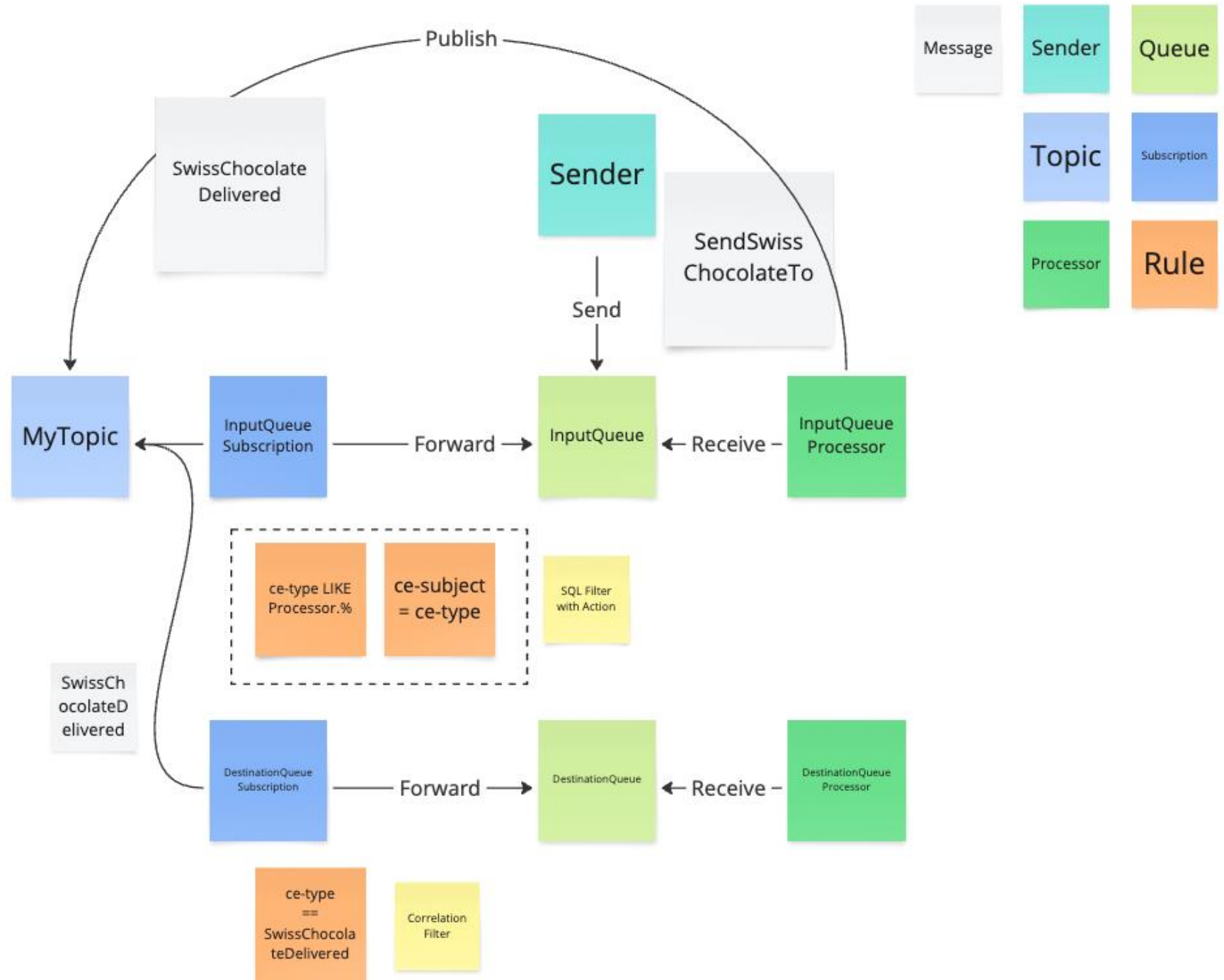
A row of various mailboxes in a grassy field with mountains in the background. The mailboxes are of different colors and styles, including white, orange, black, and rusty metal. Some have numbers like 1955, 405, 420, and 215. The background shows a clear blue sky and distant mountains.

Navigating through the **Azure Messaging** (over)choice

daniel.marbach@particular.net

Azure Service Bus

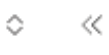






swisschocolate~events (ndcosloservicebus1/swisschocolate~events) | Subscriptions

Service Bus Topic



Subscription

Refresh

Give feedback

- Overview
- Access control (IAM)
- Diagnose and solve problems
- Service Bus Explorer
- Resource visualizer
- Settings

Name	Status	Message count	Act
swisschocolate-delivery-tracking-subscription	Active	0	0
swisschocolatewarehouse-subscription	Active	0	0



Refresh



Delete



Give feedback



Overview



Diagnose and solve problems



Service Bus Explorer



Resource visualizer



Automation



Help

Filters

A conditional expression that's evaluated in the broker against the arriving messages' user-defined properties and system properties. [Learn more](#)

+ Add filter

Name

Filter Type

SwissChocolateDelivered

CorrelationFilter



SwissChocolateDelivered

Filter Type

SQL Filter

Correlation Filter

Filter messages using a set of conditions on system and user-defined properties that must all match. [Learn more](#)

SYSTEM PROPERTIES

Key

Value

CUSTOM PROPERTIES

Key

Value

ce-type

Processor.SwissChocolateDelivered



Save changes



swisschocolatewarehouse-subscription (ndcosloservicebus1/swisschocolate~events/swisschocolatewarehouse-subs...

Service Bus Subscription



Refresh



Delete



Give feedback



Overview



Diagnose and solve problems



Service Bus Explorer



Resource visualizer



Automation



Help

Message Counts

Active

0 MESSAGES

Dead-letter

0 MESSAGES

Transfer

0 MESSAGES

Transfer dead-letter

0 MESSAGES

Filters

A conditional expression that's evaluated in the broker against the arriving messages' user-defined properties and system properties. [Learn more](#)

+ Add filter

Name

Filter Type

AllEventsPublishedUnderNamespace

SqlFilter



AllEventsPublishedUnderNamespace

Filter Type

SQL Filter

Correlation Filter

Filter messages by using a SQL-like expression that evaluates against message properties. [Learn more](#)

```
1 user.[ce-type] LIKE 'Processor.%'
```

CODE & DEMOS



-  Overview
-  Activity log
-  Access control (IAM)
-  Tags
-  Diagnose and solve problems
-  Resource visualizer
-  Events
-  Settings
 -  Geo-replication (preview)
 -  Shared access policies

Scale

-  Geo-Recovery
-  Networking
-  Encryption
-  Identity
-  Configuration
-  Properties
-  Locks
-  Entities
 -  Queues
 -  Topics
-  Monitoring
-  Automation
-  Help

Configure Run history JSON Notify Diagnostic settings

Autoscale is a built-in feature that helps applications perform their best when demand changes. You can choose to scale your resource manually to a specific messaging unit, or via a custom Autoscale policy that scales based on metric(s) thresholds, or schedule messaging unit which scales during designated time windows. Autoscale enables your resource to be performant and cost effective by adding and removing instances based on demand. [Learn more about Azure Autoscale](#) or [view the how-to video](#).

Choose how to scale your resource

 **Manual scale**
Maintain a fixed messaging unit

 **Custom autoscale**
Scale on any schedule, based on any metrics

Custom autoscale

Autoscale setting name	danielpremium-Autoscale-713		
Resource group	ReliableWorkshop		
Messaging unit	2		

Default * Auto created default scale condition 

Delete warning

The very last or default recurrence rule cannot be deleted. Instead, you can disable autoscale to turn off autoscale.

Scale mode

☒ Scale based on a metric ☐ Scale to a specific messaging unit

Rules

 It is recommended to have at least one scale in rule. To create new rules, click [Add a rule](#)

Scale out

When	danielpremium	(Average) NamespaceCpuUsag...	Increase allowed next value
Or	danielpremium	(Average) NamespaceMemory...	Increase allowed next value

+ Add a rule

Messaging unit limits

Minimum *	Maximum *	Default *
1	16	1

Schedule

This scale condition is executed when none of the other scale condition(s) match

+ Add a scale condition



Metric source

Current resource (danielpremium)

Resource type

Service Bus Namespaces

Resource

danielpremium

Criteria

Metric namespace *

Standard metrics

Metric name

CPU

1 minute time grain

No dimension is available for the selected metric.

60%

50%

40%

30%

5:37 PM

5:38 PM

5:39 PM

UTC+02:00

NamespaceCpuUsage (Average)

34 %

Operator *

Greater than

Metric threshold to trigger scale action *

60

%

Duration (minutes) *

5

Time grain (minutes)

1

Time grain statistic *

Average

Time aggregation *

Average

Action

Operation *

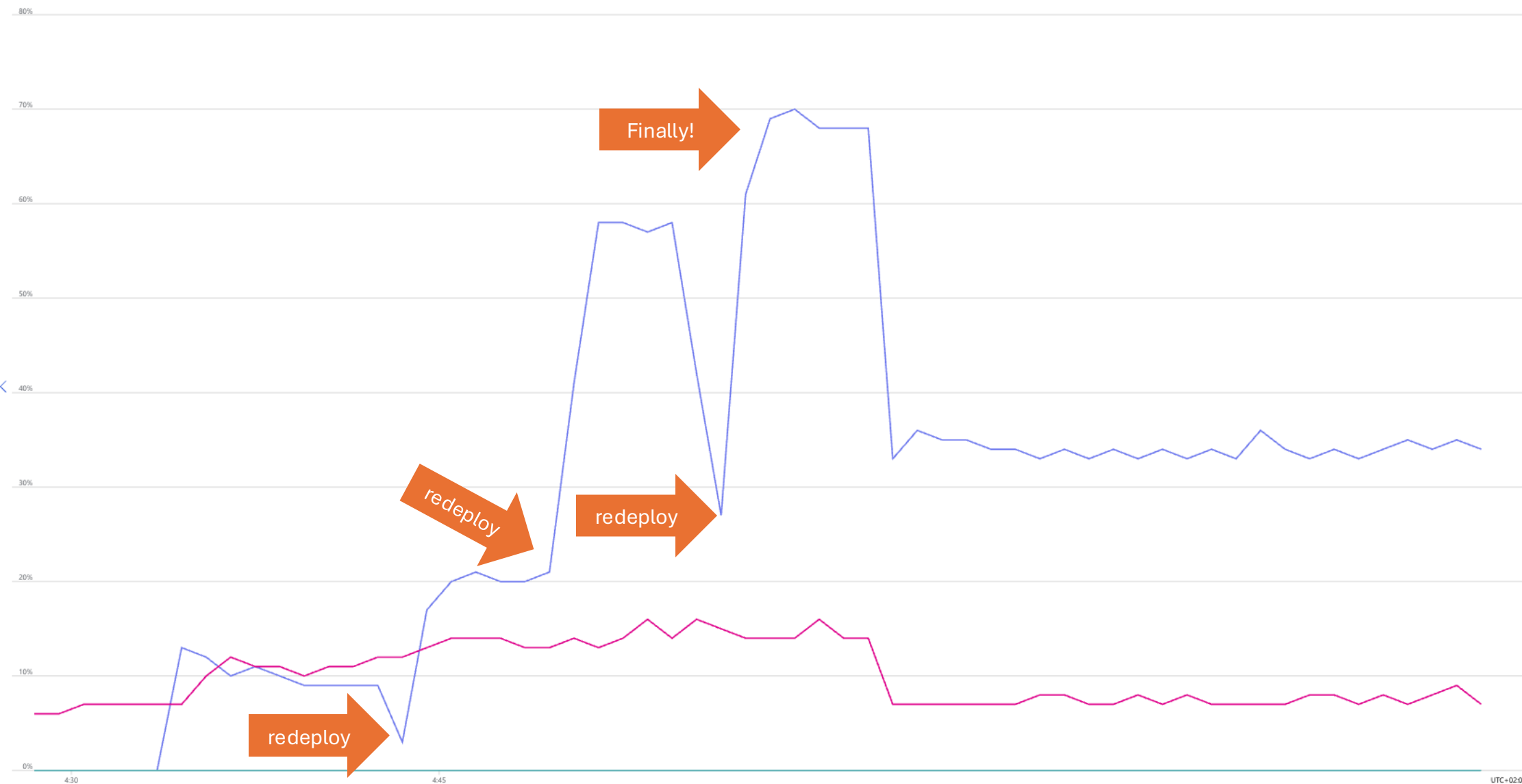
Increase

Cool down (minutes) *

5

Current messaging unit is 2; Messaging unit can scale up and down among these values [1, 2, 4, 8, 16].

[danielpremium, CPU, Avg](#) [danielpremium, Memory Usage, Avg](#) [danielpremium, Throttled Requests., Sum](#)



CPU (Avg), danielpremium | 30.1500% | Memory Usage (Avg), danielpremium | 10.0167% | Throttled Requests. (Sum), danielpremium | 0

CODE & DEMOS



Register an application ...

* Name

The user-facing display name for this application (this can be changed later).

ServiceBusRBAC



Supported account types

Who can use this application or access this API?

- ☒ Accounts in this organizational directory only (Default Directory only - Single tenant)
- ☐ Accounts in any organizational directory (Any Microsoft Entra ID tenant - Multitenant)
- ☐ Accounts in any organizational directory (Any Microsoft Entra ID tenant - Multitenant) and personal Microsoft accounts (e.g. Skype, Xbox)
- ☐ Personal Microsoft accounts only

[Help me choose...](#)

Redirect URI (optional)

We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it can be changed later, but a value is required for most authentication scenarios.

Select a platform



e.g. <https://example.com/auth>

Register an app you're working on here. Integrate gallery apps and other apps from outside your organization by adding from [Enterprise applications](#).

 ServiceBusRBAC | API permissions 

Refresh



Got feedback?



Overview



Quickstart



Integration assistant



Diagnose and solve problems



Manage



Branding & properties



Authentication



Certificates & secrets



Token configuration



API permissions



Expose an API



App roles



Owners



Roles and administrators



Manifest



> Support + Troubleshooting

Configured permissions

Applications are authorized to call APIs when they are granted permissions by users/admins as part of the consent process. [Learn more about permissions and consent](#)



Add a permission



Grant admin consent for Default Directory

API / Permissions name

Type

Description

▼ Microsoft Graph (1)

User.Read

Delegated

Sign in and read user profile

▼ Microsoft.ServiceBus (1)

user_impersonation

Delegated

Have full access to Azure Service Bus service

To view and manage consented permissions for individual apps, as well as your tenant's consent settings, try



ServiceBusRBAC | Certificates & secrets

[Got feedback?](#)[Overview](#)[Quickstart](#)[Integration assistant](#)[Diagnose and solve problems](#)[Manage](#)[Branding & properties](#)[Authentication](#)[Certificates & secrets](#)[Token configuration](#)[API permissions](#)[Expose an API](#)[App roles](#)[Owners](#)[Roles and administrators](#)[Manifest](#)[Support + Troubleshooting](#)

Credentials enable confidential applications to identify themselves to the authentic scheme). For a higher level of assurance, we recommend using a certificate (instead



Application registration certificates, secrets and federated credentials can be found

[Certificates \(0\)](#)[Client secrets \(1\)](#)[Federated credentials \(0\)](#)

A secret string that the application uses to prove its identity when requesting a to

[+ New client secret](#)

Description	Expires	Value
ServiceBusClientSecret	8/14/2025	RXk8Q~d





ServiceBusRBAC

[Delete](#)[Endpoints](#)[Preview features](#)

Overview

[Quickstart](#)[Integration assistant](#)[Diagnose and solve problems](#)

Manage

[Branding & properties](#)[Authentication](#)[Certificates & secrets](#)[Token configuration](#)[API permissions](#)[Expose an API](#)[App roles](#)[Owners](#)[Roles and administrators](#)[Manifest](#)[Support + Troubleshooting](#)

Essentials

Display name : [ServiceBusRBAC](#)

Application (client) ID : f68771ae-572

Object ID : 444d3304

Directory (tenant) ID : ffe0df17

Supported account types : [My organization only](#)

Get Started

[Documentation](#)

The Microsoft id

Add role assignment ...

Role Members Conditions Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#) ⓘ

Job function roles Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

 Azure Er



Type : All

Category : All

Name ↑↓

Description ↑↓

Azure Service Bus Data Owner

Allows for full access to Azure Service Bus resources.

Azure Service Bus Data Receiver

Allows for receive access to Azure Service Bus resources.

Azure Service Bus Data Sender

Allows for send access to Azure Service Bus resources.

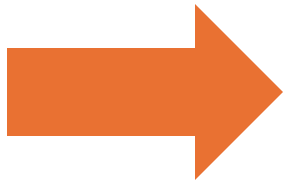
Log Analytics Reader

Log Analytics Reader can view and search all monitoring da

Monitoring Metrics Publisher

Enables publishing metrics against Azure resources

Showing 1 - 5 of 5 results.



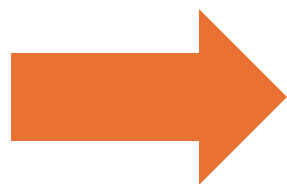
Add role assignment

Role Members Conditions Review + assign

Selected role Azure Service Bus Data Sender

Assign access to ☒ User, group, or service principal
☐ Managed identity

Members [+ Select members](#)

 Description	Name	Object ID
	ServiceBusRBAC	01426532-
	Optional	

~/Projects/AzureMessagingOverchoice/ServiceBus/ServiceBusRBAC git:(main)±17 (14.66s)

dotnet run -c Release

Using launch settings from /Users/danielmarbach/Projects/AzureMessagingOverchoice/ServiceBus/ServiceBusRBAC/Properties/launchSettings.json...

Unhandled exception. System.UnauthorizedAccessException: Unauthorized access. 'Listen' claim(s) are required to perform this operation. Resource: 'sb://ndcoslo-servicebus1.servicebus.windows.net/rbac-queue'. TrackingId:dba9c5a0083d48a0afc9e2b70fab27d8_G9, SystemTracker:gateway10, Timestamp:2025-05-16T16:01:25

For troubleshooting information, see <https://aka.ms/azsdk/net/servicebus/exceptions/troubleshoot>.

at Azure.Messaging.ServiceBus.Amqp.AmqpReceiver.ReceiveMessagesAsyncInternal(Int32 maxMessages, Nullable`1 maxWaitTime, TimeSpan timeout, CancellationToken cancellationToken)

at Azure.Messaging.ServiceBus.Amqp.AmqpReceiver.<>c.<<ReceiveMessagesAsync>b__44_0>d.MoveNext()

--- End of stack trace from previous location ---

at Azure.Messaging.ServiceBus.ServiceBusRetryPolicy.RunOperation[T1,TResult](Func`4 operation, T1 t1, TransportConnectionScope scope, CancellationToken cancellationToken, Boolean logTimeoutRetriesAsVerbose)





rbacqueue (ndcosloservicebus1/rbacqueue)

Service Bus Queue



Refresh



Delete



Give feedback



Overview



Access control (IAM)



Diagnose and solve problems



Service Bus Explorer



Resource visualizer



Settings



Automation



Help

^ Essentials

Namespace : [ndcosloservicebus1](#)

Status : [Active](#)

Partitioning : Disabled

Duplicate detection : Disabled

Dead lettering : [Disabled on message expiration](#)

^ Settings

Max delivery count
10 [\(change\)](#)

Current size
0.2 KB

Max size
1 GB [\(change\)](#)

Message time to live
14 DAYS [\(change\)](#)

^ Message Counts

Active
1 MESSAGES

Scheduled
0 MESSAGES

Dead-letter
0 MESSAGES

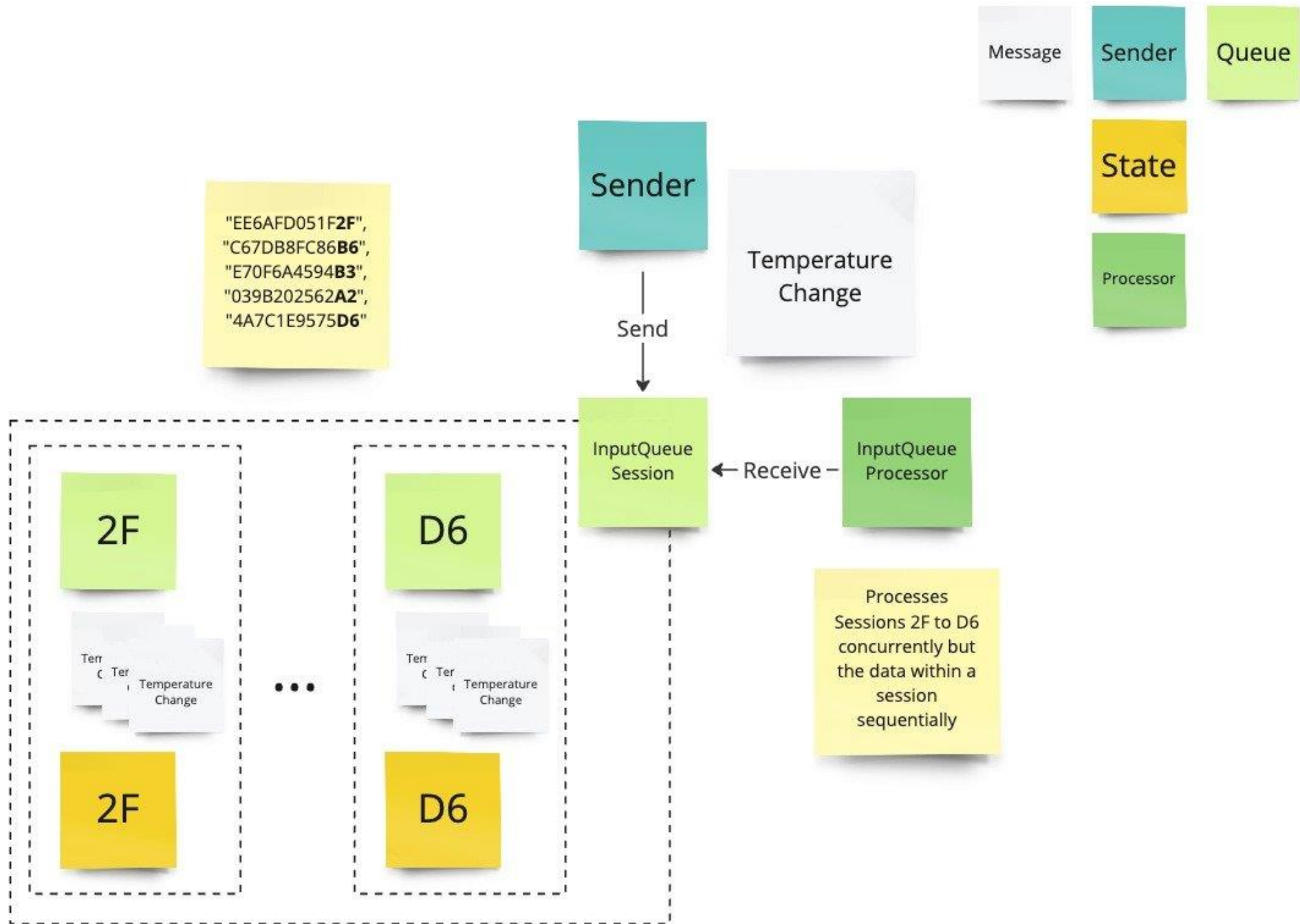
Transfer
0 MESSAGES

Transfer d
0 MESSAGES



CODE & DEMOS







chocolatestorage~temperature (ndcosloservicebus1/chocolatestorage~temperature)

Service Bus Queue



Refresh | Delete | Give feedback

Auto refresh ☒ On Refresh after 10 seconds

Overview

- Access control (IAM)
- Diagnose and solve problems
- Service Bus Explorer
- Resource visualizer
- > Settings
- > Automation
- > Help

Essentials

Namespace	: ndcosloservicebus1	Queue URL	: https://ndcosloservicebus1.servicebus.windows.net/chocolatestorage/te...
Status	: Active	Created	: Friday, May 16, 2025 at 17:26:18 GMT+2
Partitioning	: Disabled	Updated	: Friday, May 16, 2025 at 17:26:18 GMT+2
Duplicate detection	: Disabled	Sessions	: Enabled
Dead lettering	: Disabled on message expiration	Forward messages to	: Disabled



Settings

Max delivery count	Current size	Max size	Message time to live	Auto-delete	Message lock duration	Free space
10 (change)	165.4 KB	1 GB (change)	UNBOUNDED (change)	NEVER (change)	5 SECONDS (change)	100.0 %

Message Counts

Active	Scheduled	Dead-letter	Transfer	Transfer dead-letter
259 MESSAGES	0 MESSAGES	0 MESSAGES	0 MESSAGES	0 MESSAGES

Metrics

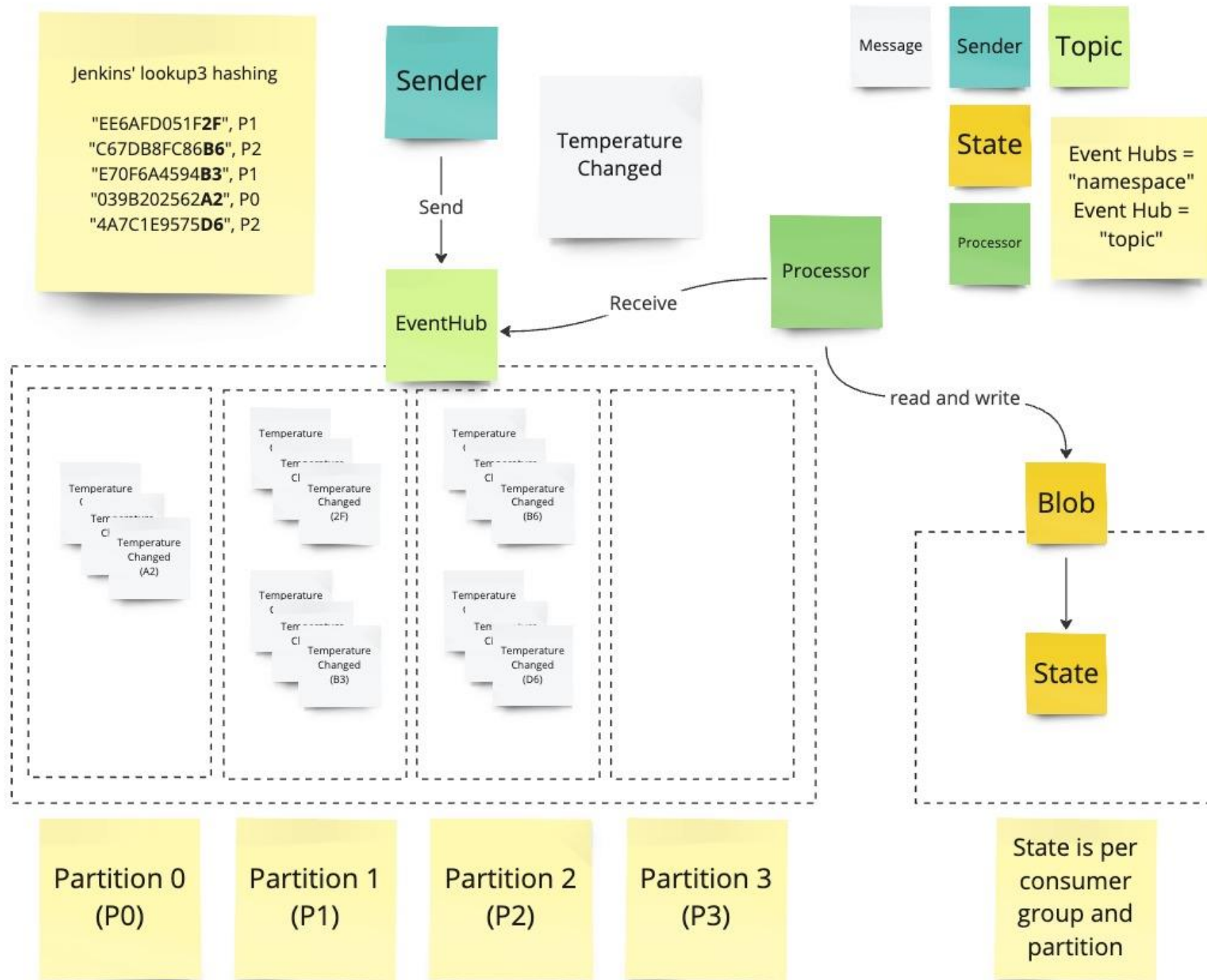
Show data for the last: 1 hour 6 hours 12 hours 1 day 7 days 30 days

Requests

Messages

Azure Event Hubs





Register an application

* Name

The user-facing display name for this application (this can be changed later).

EventHubsRBAC

Supported account types

Who can use this application or access this API?

- ☒ Accounts in this organizational directory only (Default Directory only - Single tenant)
- ☐ Accounts in any organizational directory (Any Microsoft Entra ID tenant - Multitenant)
- ☐ Accounts in any organizational directory (Any Microsoft Entra ID tenant - Multitenant) and personal Microsoft accounts (e.g. S Xbox)
- ☐ Personal Microsoft accounts only

[Help me choose...](#)

Redirect URI (optional)

We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it

[Refresh](#)[Got feedback?](#)[Overview](#)[Quickstart](#)[Integration assistant](#)[Diagnose and solve problems](#)[Manage](#)[Branding & properties](#)[Authentication](#)[Certificates & secrets](#)[Token configuration](#)[API permissions](#)[Expose an API](#)[App roles](#)[Owners](#)[Roles and administrators](#)

Configured permissions

Applications are authorized to call APIs when they are granted permissions by users/admins as part of the consent process. [Learn more about permissions and consent](#)

[Add a permission](#)[Grant admin consent for Default Directory](#)

API / Permissions name	Type	Description
▼ Microsoft Graph (1)		
User.Read	Delegated	Sign in and read user profile
▼ Microsoft.EventHubs (1)		
user_impersonation	Delegated	Have full access to the Azure Event Hub service

To view and manage consented permissions for individual apps, as well as your tenant's consent settings, try the [API permissions](#) blade.

[Refresh](#)[Got feedback?](#)[Overview](#)[Quickstart](#)[Integration assistant](#)[Diagnose and solve problems](#)[Manage](#)[Branding & properties](#)[Authentication](#)[Certificates & secrets](#)[Token configuration](#)[API permissions](#)[Expose an API](#)[App roles](#)[Owners](#)[Roles and administrators](#)

Configured permissions

Applications are authorized to call APIs when they are granted permissions by users/admins as part of the consent process. [Learn more about permissions and consent](#)

[Add a permission](#)[Grant admin consent for Default Directory](#)

API / Permissions name	Type	Description
▼ Azure Storage (1)		
user_impersonation	Delegated	Access Azure Storage
▼ Microsoft Graph (1)		
User.Read	Delegated	Sign in and read user profile
▼ Microsoft.EventHubs (1)		
user_impersonation	Delegated	Have full access to the Azure Event Hub service

To view and manage consented permissions for individual apps, as well as your tenant's consent settings, to



EventHubsRBAC | Certificates & secrets

[Got feedback?](#)[Overview](#)[Quickstart](#)[Integration assistant](#)[Diagnose and solve problems](#)[Manage](#)[Branding & properties](#)[Authentication](#)[Certificates & secrets](#)[Token configuration](#)[API permissions](#)[Expose an API](#)[App roles](#)[Owners](#)[Roles and administrators](#)

Credentials enable confidential applications to identify themselves to the authentic scheme). For a higher level of assurance, we recommend using a certificate (inste

[Certificates \(0\)](#)[Client secrets \(1\)](#)[Federated credentials \(0\)](#)

A secret string that the application uses to prove its identity when requesting a t

[New client secret](#)

Description	Expires	Value ⓘ
EventHubsClientSecret	11/14/2025	u2F8Q~





EventHubsRBAC

[Delete](#)[Endpoints](#)[Preview features](#)[Overview](#)[Quickstart](#)[Integration assistant](#)[Diagnose and solve problems](#)[Manage](#)[Branding & properties](#)[Authentication](#)[Certificates & secrets](#)[Token configuration](#)[API permissions](#)

^ Essentials

Display name : [EventHubsRBAC](#)

Application (client) ID : 5a02a272-f0e1

Object ID : 5c090ee8-6e28

Directory (tenant) ID : ffe0df17-c

Supported account types : [My organization only](#)

[Get Started](#)[Documentation](#)

Add role assignment ...

Role **Members** Conditions Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#) 

Job function roles Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

 Data owner



Type : **All**

Category : **All**

Name 

Description 

Azure Event Hubs Data Owner

Allows for full access to Azure Event Hubs resources.

Showing 1 - 1 of 1 results.



Add role assignment ...

[Role](#) [Members](#) [Conditions](#) [Review + assign](#)

Selected role Azure Event Hubs Data Owner

Assign access to ☒ User, group, or service principal
 ☐ Managed identity

Members [+ Select members](#)



Name	Object ID
EventHubsRBAC	196974c4-

Description Optional

Add role assignment

Role **Members** Conditions Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Job function roles Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.



Type : **All**

Category : **All**

Name ↑↓

Description ↑↓

Schema Registry Contributor

Read, write, and delete Schema Registry groups and schemas.

Schema Registry Reader

Read and list Schema Registry groups and schemas.

Showing 1 - 2 of 2 results.



Add role assignment

[Role](#) [Members](#) [Conditions](#) [Review + assign](#)

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

[Job function roles](#) [Privileged administrator roles](#)

Grant access to Azure resources based on job function, such as the ability to create virtual machines.



Type : All

Category : All

Name ↑↓

Description ↑↓

Defender CSPM Storage Data Scanner

Grants access to read blobs and files. This role is used by the data scanner of Dfender CSPM.

Defender for Storage Data Scanner

Grants access to read blobs and update index tags. This role is used by the data scanner of Defender for Storage.

Storage Blob Data Contributor

Allows for read, write and delete access to Azure Storage blob containers and data

Storage Blob Data Owner

Allows for full access to Azure Storage blob containers and data, including assigning POSIX access control.

Storage Blob Data Reader

Allows for read access to Azure Storage blob containers and data

Storage Blob Delegator

Allows for generation of a user delegation key which can be used to sign SAS tokens





Schemas (ndcosloeventhub1/Schemas)



Schema Group



Schema



Delete



Refresh



Give feedback



Overview



Diagnose and solve problems



Resource visualizer



Settings



Locks



Automation



Help



Essentials

Resource group ([move](#)) : [NdcOslo](#)

Location : North Europe

Subscription ([move](#)) : [Microsoft Azure Sponsorship](#)

Subscription ID : 70931878-44ae-4019-bdfe-f2401f97119d

Namespace : [ndcosloeventhub1](#)



Schemas define the contract between the producers and consumers. Schemas defined in Event Hubs Schema Registry helps manage the schema evolution.

Schemas (0)



Name	ID
------	----

Create Schema

Schema

Name * ⓘ

Processor.StorageTemperatureChanged

Upload schema here ⓘ

Select a file

```
1  {
2    "namespace": "Processor",
3    "type": "record",
4    "name": "StorageTemperatureChanged",
5    "fields": [
6      {
7        "name": "Current",
8        "type": "double"
9      },
10     {
11       "name": "Published",
12       "type": {
13         "type": "int",
14         "logicalType": "date"
15       }
16     }
17   ]
18 }
```



ndcosloeventhub1 | Application Groups



Event Hubs Namespace | Application Group



Application Group



Refresh



Overview



Activity log



Access control (IAM)



Tags



Diagnose and solve problems



Data Explorer



Resource visualizer



Events



Settings



Shared access policies



Scale



Geo-Recovery



Networking



Encryption



Identity



Configuration



Application Groups



Properties



Locks



With application groups, you can create logi



Search to filter items by name...

Name

No results.



Add application group

Application Group

Name *



EventHubsRBAC



Enabled



Security context type

Microsoft Entra application



Microsoft Entra Application (client) ID *



5a02a272-



[Get Microsoft Entra Application \(client\) ID](#)



Click "Get Microsoft Entra Application (client) ID" above to open the Microsoft Entra applications page. Copy the "Application (Client) ID" of the desired app, return to this page, and paste it into the textbox above.

Client group ID



AADAppID=5a02a272-



Policies

Add to the list of group policies that define the behavior of this application group. The policies can support resource governance scenarios such as limiting ingress or egress traffic.

Name	Type	Metric ID	Rate limit threshold
Throttle	ThrottlingPolicy	IncomingMessages	1
The name of this policy	Policy type	Metric Id on which...	Rate limit per second

info: Azure.Messaging.EventHubs[3]

Publishing events for Event Hub: temperaturechanged (Partition Id/Key: 'EE6AFD051F2F'), Operation Id: 'e664231a-f2e4-4903-8f53-6c2d3a7f8cd8'.

fail: Azure.Messaging.EventHubs[5]

An exception occurred while publishing events for Event Hub: temperaturechanged (Partition Id/Key: 'EE6AFD051F2F'), Operation Id: 'e664231a-f2e4-4903-8f53-6c2d3a7f8cd8'. Error Message: 'The request was terminated because the entity is being throttled by the application group with application group name EventHubsRBAC and policy name Throttle.Error code: 50013.S:N:NDCOSLOEVENTHUB1:EVENTHUB:TEMPERATURECHANGED~8191,CL:1000000,CC:2040,ACC:80010,LUR:,LUT:,RC:2040 TrackingId:7da178fe000019be00000d5682a3a22_G10_B28, SystemTracker:ndcosloeeventhub1:eventhub:temperaturechanged~8191, Timestamp:2025-05-18T19:51:03 (temperaturechanged).

For troubleshooting information, see <https://aka.ms/azsdk/net/eventhubs/exceptions/troubleshoot>'

info: Azure.Messaging.EventHubs[3]

Publishing events for Event Hub: temperaturechanged (Partition Id/Key: 'EE6AFD051F2F'), Operation Id: 'e664231a-f2e4-4903-8f53-6c2d3a7f8cd8'.

fail: Azure.Messaging.EventHubs[5]

An exception occurred while publishing events for Event Hub: temperaturechanged (Partition Id/Key: 'EE6AFD051F2F'), Operation Id: 'e664231a-f2e4-4903-8f53-6c2d3a7f8cd8'. Error Message: 'The request was terminated because the entity is being throttled by the application group with application group name EventHubsRBAC and policy name Throttle.Error code: 50013.S:N:NDCOSLOEVENTHUB1:EVENTHUB:TEMPERATURECHANGED~8191,CL:1000000,CC:0,ACC:82050,LUR:,LUT:,RC:2040 TrackingId:7da178fe000019be00000d5682a3a22_G10_B28, SystemTracker:ndcosloeeventhub1:eventhub:temperaturechanged~8191, Timestamp:2025-05-18T19:51:15 (temperaturechanged). For troubleshooting information, see <https://aka.ms/azsdk/net/eventhubs/exceptions/troubleshoot>'

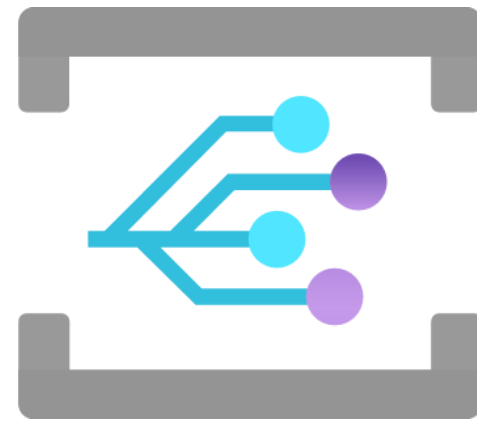
info: Azure.Messaging.EventHubs[3]

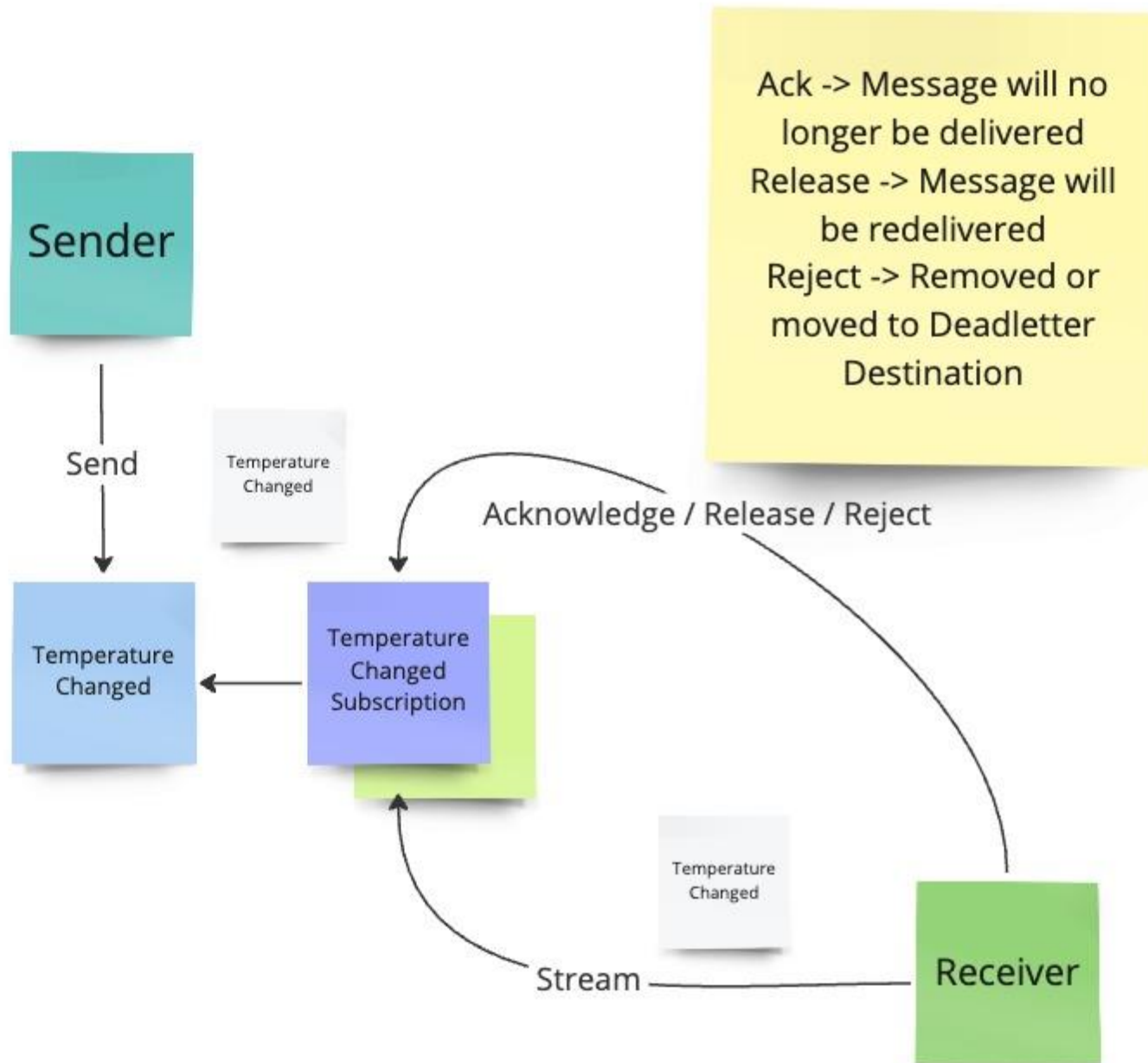
Publishing events for Event Hub: temperaturechanged (Partition Id/Key: 'EE6AFD051F2F'), Operation Id: 'e664231a-f2e4-4903-8f53-6c2d3a7f8cd8'.

CODE & DEMOS



Azure Event Grid





 ndcosloeventgridstorage | Events ☆ ...
Storage account

 Search
 

 Event Subscription
  Refresh
  Give feedback

- Overview
- Activity log
- Tags
- Diagnose and solve problems
- Access Control (IAM)
- Data migration
- Events**
- Storage browser
- Storage Mover
- Partner solutions
- Resource visualizer
- > Data storage
- ✓ Security + networking
 - Networking
 - Front Door and CDN
 - Access keys
 - Shared access signature
 - Encryption
 - Microsoft Defender for Cloud

[Get Started](#) [Event Subscriptions](#)

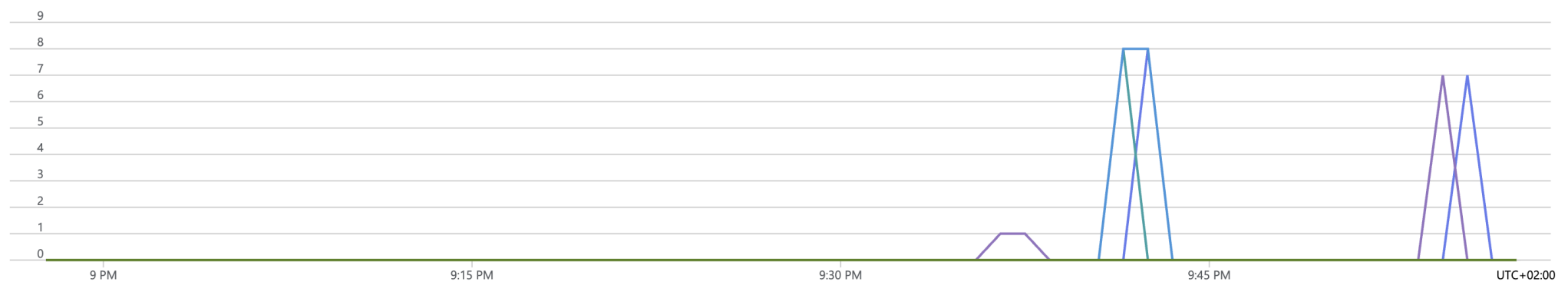
^ Essentials

System Topic : [ndcosloeventgridstoragesystemtopic](#)

Topic Type : Microsoft.Storage.StorageAccounts

Show metrics: **General** Errors Latency Dead-Letter

For the last: **1 hour** 6 hours 12 hours 1 day 7 days 30 days



Published Events (Sum), ndcosloeventgridstoragesystemtopic 17	Publish Failed Events (Sum), ndcosloeventgridstoragesystemtopic --	Matched Events (Sum), ndcosloeventgridstoragesystemtopic 17
Delivered Events (Sum), ndcosloeventgridstoragesystemtopic 9	Dead Lettered Events (Sum), ndcosloeventgridstoragesystemtopic --	Delivery Failed Events (Sum), ndcosloeventgridstoragesystemtopic 16
Dropped Events (Sum), ndcosloeventgridstoragesystemtopic --	Advanced Filter Evaluations (Sum), ndcosloeventgridstoragesystemtopic --	

Name	Endpoint	Prefix Filter	Suffix Filter	Event Types	Provisioning state
 subscription	NamespaceTopic			Microsoft.Storage.BlobCreated,Microsoft.Storage.Blob...	Succeeded

EVENT TYPES

Filter to Event Types

Event Schema

SUBJECT FILTERS

Apply filters to the subject of each event. Only events with matching subject will be returned.

☐ Enable subject filtering

ADVANCED FILTERS

Filter on attributes of each event. Only events that match all filters will be returned. Case-insensitive. [Learn more](#)

Valid keys for currently selected event schema:

- id, source, type, specversion
- Custom properties, using "." as the nesting separator. (e.g. "my.custom.property")

Key	Operator
-----	----------

No results

[Add new filter](#)

☒ Enable advanced filtering on arrays ⓘ

2 selected

☒ Blob Created

☒ Blob Deleted

☐ Directory Created

☐ Directory Deleted

☐ Blob Renamed

☐ Directory Renamed

☐ Blob Tier Changed

☐ Blob Inventory Completed

☐ Async Operation Initiated

☐ Lifecycle Policy Completed

 subscription ...

Event Subscription

 Save

 Delete

 Give feedback

TOPIC



Storage account
[ndcosloeventgridstorage](#)

Metrics

Filters

Additional Features

Delivery Properties

MANAGED IDENTITY FOR DELIVERY

Managed identities are used to authenticate an Event Grid topic to Azure service instances when delivering events. Select either a system assigned or a user assigned managed identity. You should have already configured one or more identities on the topic to which this event subscription is associated. [Learn more about Managed Identities](#)

Managed identity type

System Assigned

▼

 ndcosloeventgridtopic (ndcosloeventgrid/ndcosloeventgridtopic) ☆ ⋮

Event Grid Namespace Topic

 Search

◇

⏪

 Overview

-  Activity log
-  Access control (IAM)
-  Resource visualizer
- ▼ Settings
 -  Access keys
 -  Properties
 -  Locks
- ▼ Entities
 -  Subscriptions
 - > Automation
 - > Help

 Subscription

 Refresh

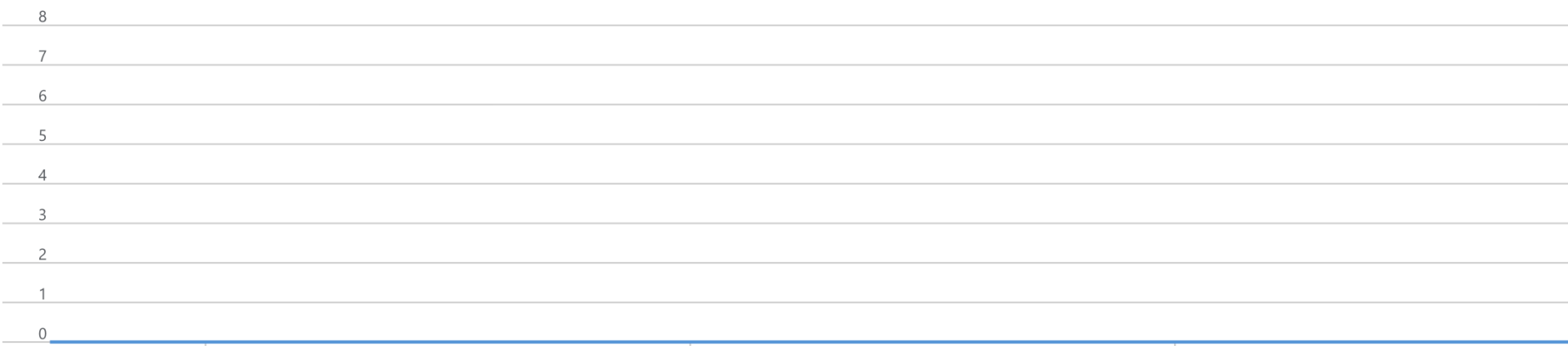
 Delete

 Give feedback

^ Essentials

Namespace	: ndcosloeventgrid	Provisioning state	: Succeeded
Resource group	: NdcOslo	Publisher type	: Custom
Subscription	: Microsoft Azure Sponsorship	Input schema	: CloudEventSchemaV1_0
Subscription ID	: 547ecb3d-ee58-447d-8ebe-9e34f4c03adb	Retention in days	: 7

Chart type ☒ General ☐ Errors ☐ Latency



 ndcosloeventgridtopic (ndcosloeventgrid/ndcosloeventgridtopic) | Subscriptions ☆ ⋮

Event Grid Namespace Topic

◇

⏪

-  Overview
-  Activity log
-  Access control (IAM)
-  Resource visualizer
- ▼ Settings
 -  Access keys
 -  Properties
 -  Locks
- ▼ Entities
-  **Subscriptions** ☆
- > Automation
- > Help

Subscriptions

Use event subscriptions to select the events on the associated topic that should be delivered to a destination or client. [Learn more](#)

 Subscription  Refresh  Give feedback

Name	Delivery mode	Delivery schema
ndcosloeventgridtopicsubscription	Queue	CloudEventSchemaV1_0



Search



Add



Download role assignments



Edit columns



Refresh



Delete



Feedback



Overview



Activity log



Access control (IAM)



Resource visualizer



Settings



Access keys



Properties



Locks



Entities



Subscriptions



Automation



Help

Check access

Role assignments

Roles

Deny assignments

Classic administrators

Number of role assignments for this subscription ⓘ

9

4000



Search by name or email

Type : All

Role : All

Scope : All scopes

Group by : Role

All (3)

Job function roles (1)

Privileged administrator roles (2)



Name ↑↓

Type ↑↓

Role ↑↓

Scope ↑↓



EventGrid Data Sender (1)



ndcosloeventgridstoragesystemt...
3c032290-476a-478b-a5da-738...

Managed identity

EventGrid Data Sender

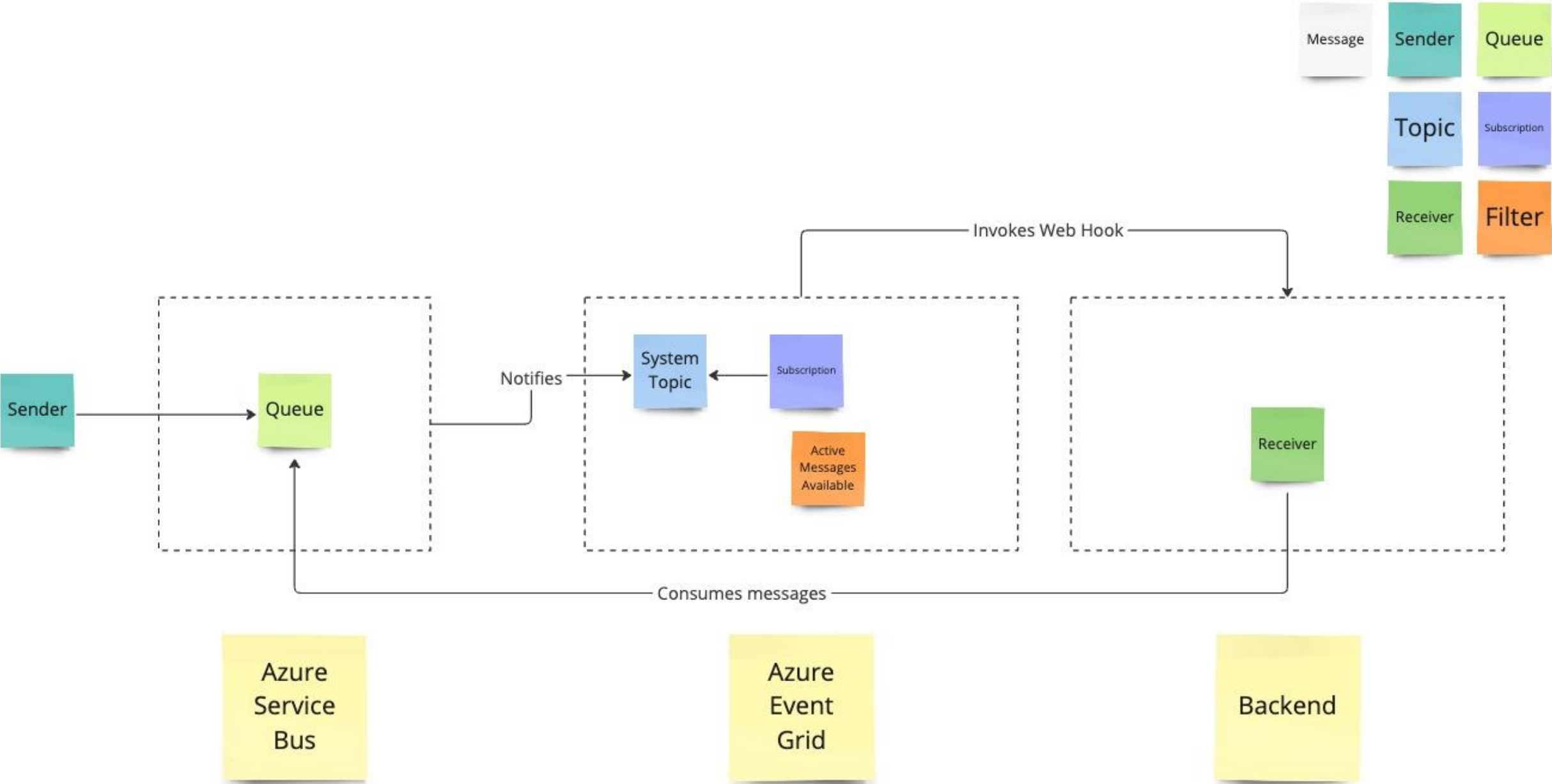


This resource

Showing 1 - 1 of 1 results.

CODE & DEMOS





 Search



 Event Subscription

 Refresh

 Give feedback

-  Overview
-  Activity log
-  Access control (IAM)
-  Tags
-  Diagnose and solve problems
-  Resource visualizer
-  **Events**
- Settings

 Geo-replication (preview)

 Shared access policies

 Scale

 Geo-Recovery

 Networking

 Encryption

 Identity

 Configuration

 Properties

 Locks

Entities

Get Started

Event Subscriptions

^

Essentials

System Topic : [ndcosloeventgridservicebussystemtopic](#)

Topic Type : Microsoft.ServiceBus.Namespaces

Show metrics:

General

Errors

Latency

Dead-Letter

For the last:

1 hour

6 hours

12 hours

1 d



-  Published Events (Sum), ndcosloeventgridservicebussystemtopic | --

 Publish Failed Events (Sum), ndcosloeventgridservicebussystemtopic | --

 Matched Events (Sum), ndcosloeventgridservicebussystemtopic | --
-  Delivered Events (Sum), ndcosloeventgridservicebussystemtopic | --

 Dead Lettered Events (Sum), ndcosloeventgridservicebussystemtopic | --

 Delivery Failed Events (Sum), ndcosloeventgridservicebussystemtopic | --
-  Dropped Events (Sum), ndcosloeventgridservicebussystemtopic | --

 Advanced Filter Evaluations (Sum), ndcosloeventgridservicebussystemtopic | --

 Search to find event subscription by name...

Name	Endpoint	Prefix Filter	Suffix Filter	Event Types
 subscription	WebHook			Microsoft.ServiceBus.ActiveMessagesAvailableWithNo...

TOPIC



Service Bus Namespace
[ndcosloervicebus1prem](#)

Metrics Filters Additional Features Delivery Properties

Show metrics:

General Errors Latency

For the last:

1 hour 6 hours 12 hours 1 day 7 days 30 days



Delivered Events (Sum), ndcosloeventgridservicebussystemtopic | 0

Delivery Failed Events (Sum), ndcosloeventgridservicebussystemtopic | --

Matched Events (Sum), ndcosloeventgridservicebussystemtopic | --

Dropped Events (Sum), ndcosloeventgridservicebussystemtopic | --

Advanced Filter Evaluations (Sum), ndcosloeventgridservicebussystemtopic | --

Web Hook [\(change\)](#)

<https://jolly-lake-3qnmzg1.euw.devtnnels.ms:8080/api/EventGridEventHandler> [\(change\)](#)



TOPIC



Service Bus Namespace
ndcosloservicebus1prem

Metrics **Filters** Additional Features Delivery Properties

EVENT TYPES

Filter to Event Types

ActiveMessagesAvailableWithNoListeners

Event Schema

CloudEventSchemaV1_0

SUBJECT FILTERS

Apply filters to the subject of each event. Only events with matching subjects get delivered. [Learn more](#)

☐ Enable subject filtering

ADVANCED FILTERS

Filter on attributes of each event. Only events that match all filters get delivered. Up to 25 filters can be specified. All string comparisons are case-insensitive. [Learn more](#)

Valid keys for currently selected event schema:

- id, source, type, specversion
- Custom properties, using "." as the nesting separator. (e.g. key, key1.key2)

Key	Operator	Value
data.QueueName	String is in	queue   Add new value (up to 25) 

[Add new filter](#)

☒ Enable advanced filtering on arrays ⓘ

TOPIC



Service Bus Namespace
[ndcosloservicebus1prem](#)

Metrics

Filters

Additional Features

Delivery Properties

PROPERTIES SENT ON EVENT DELIVERY

Define headers that are included with the request sent to the destination. For example, you may want to set a fixed value for the Authorization header to be sent with the request. Alternatively, you can set the value as a JsonPath reference to an existing envelope property or to any property in the event data. Only JSON values of string, number and boolean are supported. For more information, consult [Event Grid documentation](#).

Valid dynamic header source fields for currently selected event schema:

- id, source, type, specversion
- Custom properties, using "." as the nesting separator. (e.g. key, key1.key2)

Header name	Type	Value	Secret	
X-Tunnel-Authorization	Static	tunnel eyJhbGciOiJFUzI1...	True	
	Static		☐ Is secret?	

→ PushDelivery git:(main) ✖ dotnet run -c Release

info: Microsoft.Hosting.Lifetime[14]

Now listening on: http://localhost:8080

info: Microsoft.Hosting.Lifetime[0]

Application started. Press Ctrl+C to shut down.

info: Microsoft.Hosting.Lifetime[0]

Hosting environment: Production

info: Microsoft.Hosting.Lifetime[0]

Content root path: /Users/danielmarbach/Projects/AzureMessagingOverchoice/EventGrid/PushDelivery

info: System.Net.Http.HttpClient.Default.LogicalHandler[100]

Start processing HTTP request GET https://rp-northeurope.eventgrid.azure.net:553/eventssubscriptions/subscription/validate?id=7737B9E1-007A-41CF-BFC7-A7C377D77625&t=2025-05-19T20:48:14.8970735Z&apiVersion=2023-06-01-preview&token=aUN12l8k5q1Kv3FVERq0zrlgkjD%2fgTX9%2bf2zotuCjk0%3d

info: System.Net.Http.HttpClient.Default.ClientHandler[100]

Sending HTTP request GET https://rp-northeurope.eventgrid.azure.net:553/eventssubscriptions/subscription/validate?id=7737B9E1-007A-41CF-BFC7-A7C377D77625&t=2025-05-19T20:48:14.8970735Z&apiVersion=2023-06-01-preview&token=aUN12l8k5q1Kv3FVERq0zrlgkjD%2fgTX9%2bf2zotuCjk0%3d

info: System.Net.Http.HttpClient.Default.ClientHandler[101]

Received HTTP response headers after 371.3546ms - 200

info: System.Net.Http.HttpClient.Default.LogicalHandler[101]

End processing HTTP request after 380.3682ms - 200

CODE & DEMOS



Segment	Eventing PubSub	Streaming	Enterprise Messaging
Product	Event Grid	Event Hubs	Service Bus
What do you care about	• Communication between apps / orgs • Individual message • Push and pull semantics • Filtering and routing • Pay as you go • Fan out	• Many messages in a Stream (think in MBs) • Ease of use and operation • Low cost • Fan in • Strict ordering • Kafka compliant • Works with other tools	• Instantaneous consistency • Strict ordering • JMS • Non-repudiation & Security • Geo-Replication & Availability • Rich features (de-dupe, scheduling, etc.)
What you care less about	• Ordering of messaging • Instantaneous consistency	• Individual message semantics • Server-side cursor • Only Once	• Cost • Simplicity
Focus point	• Reliable fan out push at massive scale	• High scale distributed log	• Highly indexed full featured message broker
	Serverless	Streaming Data	Enterprise



Use all of them



Filter by title

Service Bus Messaging Documentation

> Overview

> Quickstarts

> Tutorials

Update inventory

> Handle Service Bus events via Event Grid

Build message-driven business applications with
NServiceBus

> Samples

> Concepts

> How-to guides

> Reference

> Resources



Learn / Azure / Messaging services / Service Bus Messaging /



Build message-driven business applications with NServiceBus and Azure Service Bus

Article • 06/21/2024 • 4 contributors

Feedback

go.particular.net/
ndc-oslo-2025-azure-apps

NServiceBus is a commercial messaging framework provided by Particular Software. It's built on top of Azure Service Bus and helps developers focus on business logic by abstracting infrastructure concerns. In this guide, we'll build a solution that exchanges messages between two services. We'll also show how to automatically retry failing messages and review options for hosting these services in Azure.

Note

The code for this tutorial is available on the [Particular Software Docs web site](#).



[github.com/danielmarbach/
AzureMessagingOverchoice](https://github.com/danielmarbach/AzureMessagingOverchoice)



A row of various mailboxes in a grassy field with mountains in the background. The mailboxes are of different colors and styles, including white, orange, black, and rusty metal. Some have numbers like 1955, 405, 420, and 215. The background shows a clear blue sky and distant mountains.

Navigating through the **Azure Messaging** (over)choice

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